

Python 3 Complete Master Guide

1. Introduction & Basic Syntax

Python is a high-level, interpreted, dynamically typed programming language known for clean code readability.

Key Features:

- Simple, expressive syntax
- Dynamically typed with optional type hints
- Vast standard library and PyPI ecosystem

2. Core Data Types

Primitives: int, float, str, bool, bytes

Collections: list (mutable sequence), tuple (immutable sequence), dict (key-value mapping), set (unique unordered collection)

3. Control Flow & Functions

Conditionals: if, elif, else

Loops: for, while, break, continue

Functions: defined with 'def', supports *args, **kwargs, lambda expressions, and type annotations.